Categorization of Malignant and Benign Breast Cancer Cells using Decision Trees: A Comparison Between Different Models

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Abstract

Cancer remains one of the most formidable challenges in contemporary medicine, with early and accurate diagnosis being pivotal in effective treatment. This paper explores the application of decision trees, a popular machine learning algorithm, for the identification of malignant cancer cells. Decision trees offer a transparent and interpretable model for classifying cancer cells, making them highly suitable for clinical decision support systems. The study provides a comprehensive comparison of two decision tree algorithms - REPTree and J48. Confusion matrices of the classification results were analyzed to compare the effectiveness of the two different algorithms. The two decision tree models were then compared to other algorithms - random forest and logistic regression. Results showed that the J48 algorithm outperformed the REPTree algorithm, but both decision trees underperformed the random forest and logistic regression models.

Introduction

Cancer cells can be categorized as either malignant or benign and being able to distinguish between the two is crucial for diagnosing patients with cancer. Physical features of the cell can be used to predict which category a given cancer cell belongs to - malignant (M) or benign (B). In this paper we summarize how different decision tree algorithms, logistic regression and random forest perform in this categorization problem.

Methods & Results

Data was obtained from an open source repository from Kaggle and decision tree, logistic regression and random forest were performed using Weka software.

Data Overview

The data had a total of 32 attributes. The data had the following attributes as summarized in the table below:

Table 1: List and Type of Attributes for Cancer Dataset

Attribute Description	Data Type
id	Integer
diagnosis	Character
radius_mean	Numeric
texture_mean	Numeric
perimeter_mean	Numeric
area_mean	Numeric

smoothness_mean	Numeric
compactness_mean	Numeric
concavity_mean	Numeric
concave.points_mean	Numeric
symmetry_mean	Numeric
fractal_dimension_mean	Numeric
radius_se	Numeric
texture_se	Numeric
perimeter_se	Numeric
area_se	Numeric
smoothness_se	Numeric
compactness_se	Numeric
concavity_se	Numeric
concave.points_se	Numeric
symmetry_se	Numeric
fractal_dimension_se	Numeric
radius_worst	Numeric
texture_worst	Numeric
perimeter_worst	Numeric
area_worst	Numeric
smoothness_worst	Numeric
compactness_worst	Numeric
concavity_worst	Numeric
concave.points_worst	Numeric
symmetry_worst	Numeric
fractal_dimension_worst	Numeric

Decision Tree with REPTree Algorithm

All decision trees were run using WEKA software. Decision trees were run using the REPTree and J48 decision tree algorithms with 10k folds cross-validation.

The first tree ran using REPTree using default parameters.

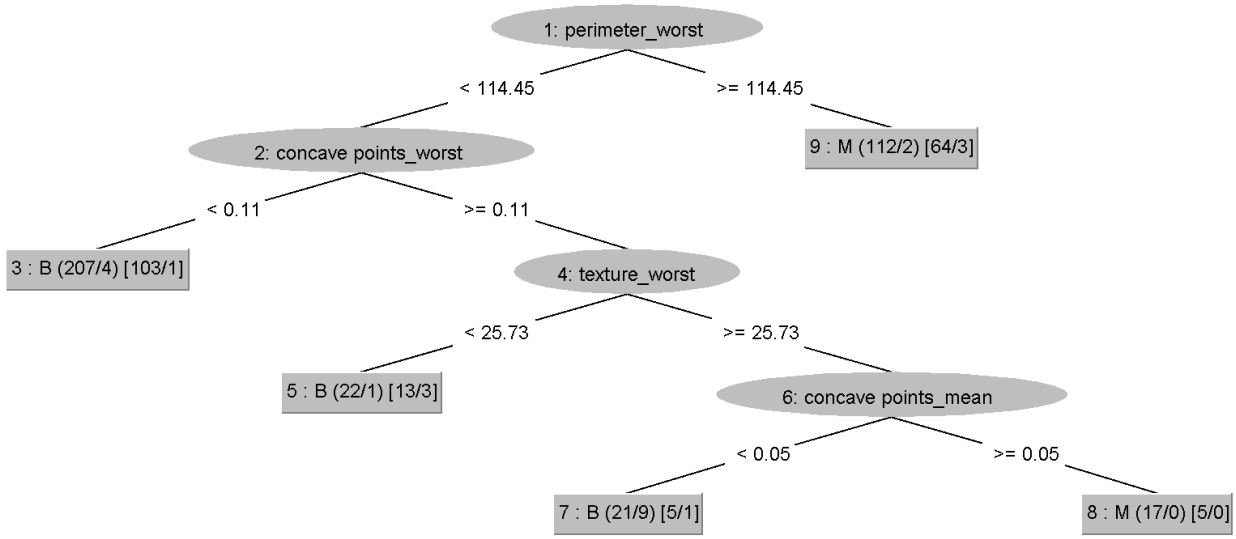


Figure 1 Decision Tree 10k Folds using REPTree with default settings (Depth = 5)

=== Confusion Matrix ===

```
  a   b   <-- classified as
188  24 |   a = M
 19 338 |   b = B
```

Decision Tree with J48 Algorithm

Another decision tree was used using the default settings of the J48 algorithm.

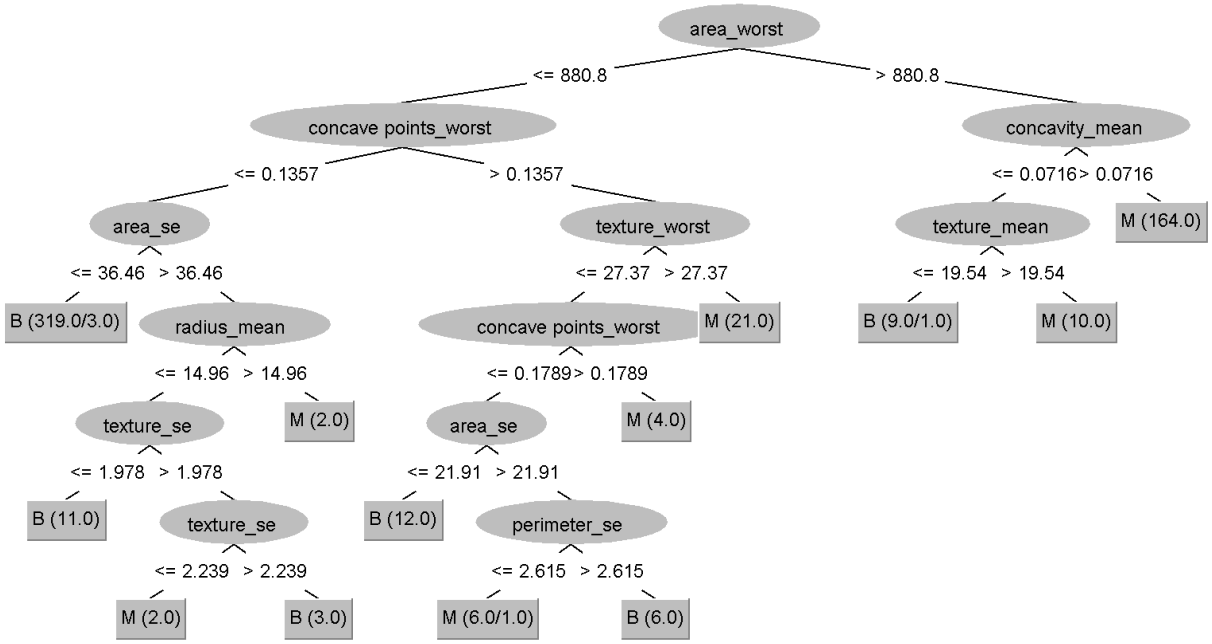


Figure 2 J48 Decision Tree

=== Confusion Matrix ===

```
  a   b   <-- classified as
195  17 |   a = M
 22 335 |   b = B
```

Decision Tree with J48 Algorithm with Pruning

The decision tree with the J48 algorithm was run again, but this time pruning was enabled

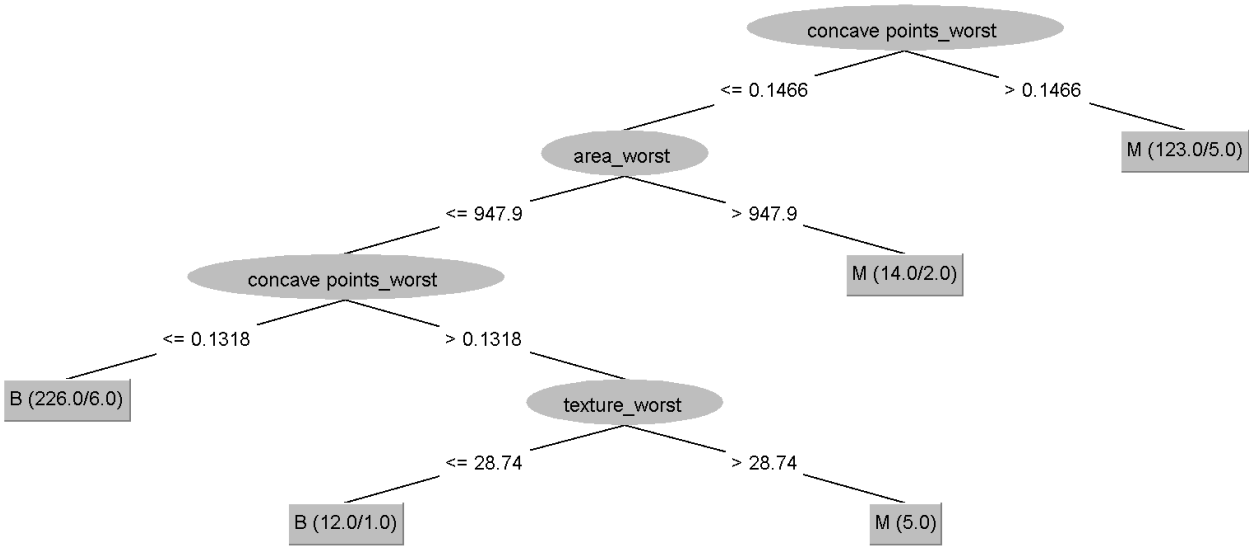


Figure 3 J48 Decision Tree (Pruned)

=== Confusion Matrix ===

```
  a   b   <-- classified as
194  18 |   a = M
 18 339 |   b = B
```

The data was then analyzed with Random Forests and logistic regressions and the confusion matrices of the two methods are shown below:

Random Forest

=== Confusion Matrix ===

```
  a   b   <-- classified as
196  16 |   a = M
  7 350 |   b = B
```

Logistic Regression

=== Confusion Matrix ===

```
  a   b   <-- classified as
200  12 |   a = M
 21 336 |   b = B
```

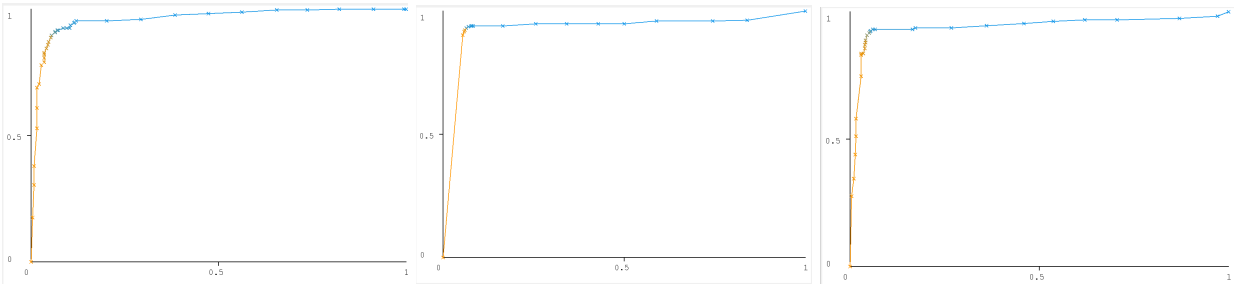
	REPTree	**J48**	**J48(Pruned)**
Size	5	13	5
# of Leaves	9	25	9
# of Unique Attributes in Model	4	12	3

Table 2: Decision Tree Statistics

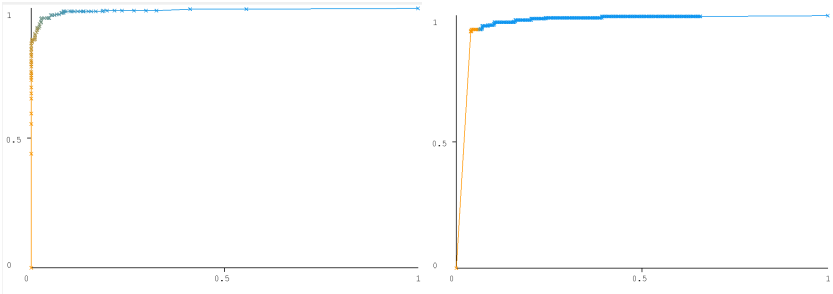
	REPTree	**J48**	**J48(Pruned)**	**Random Forest**	**Logistic Regression**
Correctly Classified Instances	92.4429%	93.1459%	93.6731%	95.9578%	94.2004%
ROC Curve	0.92	0.93	0.94	0.99	0.97

Table 3: Performance Comparison Decision Trees

Figure 4: ROC Curve



REPTree **J48** **J48 Pruned**



Random Forest **Logistic Regression**

Discussion

In terms of performance, between the J48 algorithm and the REPTree algorithm, the J48 has greater accuracy. The J48 algorithm however has a much more complex tree with a size of 25 and 13 leaves versus a size of 9 and 4 leaves with REPTree.

By setting the prune setting to true and allowing the tree to be pruned, we were able to come up with a simpler model using the J48 algorithm which reduced the tree size and depth while roughly maintaining the same accuracy. Our tree with this modified J48 algorithm reduces the size to 9 and the number of leaves to 5. Only 3 attributes were used by the tree (concave points_worst, area_worst, and texture_worst) as opposed to 12 from the unpruned tree. This means that with a simpler tree model using fewer attributes can be used without sacrificing too much accuracy.

The ROC curves for the J48 algorithm, J48 algorithm with pruning and the REPTree algorithm show that the performance can be ranked as follows: J48 > J48 (Pruned) -> REPTree. This is reflected by the fact that the ROC area is greatest for J48 and smallest for REPTree among the three decision trees we constructed.

All three of our decision trees were outperformed by both logistic regression and random forest models. This is once again reflected in the ROC curves which are higher and closer to the top left corner for the logistic and random forest models compared to the decision tree models. The ROC area values also reflect this fact since the areas are greater for random forest and logistic regression versus the decision trees.

In summary, we have successfully demonstrated how to use different decision tree algorithms and parameters in Weka to solve a categorization problem in medicine, and compared them to other categorization methods.